

3.5.3 Nested Loops and for vs while

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ ش...

CS XI

Chapter 3: Programming Fundamentals

Lecture 19 of 24

★ Golden

3.5.3 Nested Loops and for vs while ★ Golden

LEARN IT

A **nested loop** is a loop inside another loop. The **inner loop runs completely, from start to finish, for every single iteration of the outer loop**. If the outer loop runs 2 times and the inner loop runs 3 times, the inner body runs $2 \times 3 = 6$ times.

```
for outer in range(1, 3):      # runs 2 times (1, 2)
    for inner in range(1, 4):  # runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
```

OUTPUT

```
Outer: 1, Inner: 1
Outer: 1, Inner: 2
Outer: 1, Inner: 3
Outer: 2, Inner: 1
Outer: 2, Inner: 2
Outer: 2, Inner: 3
```

WORKED EXAMPLE

A **star triangle**. The outer loop chooses the row; the inner loop prints that many stars on the same line (`end=""`), and the empty `print()` moves to the next line.

```
for row in range(1, 5):
    for col in range(row):
        print("*", end="")
    print()
```

OUTPUT

```
*
**
***
****
```

Row 1: inner loop runs once → *. Row 2: twice → **. And so on up to row 4.

for loop vs while loop

for loop	while loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the loop variable.	Requires a manual update of the counter (risk of an infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic situations (like "keep playing until the user says Stop").
<code>for i in range(1, 4): print(i)</code>	<code>i = 1 / while i ≤ 3: / print(i) / i += 1</code>

REMEMBER IT

Ask yourself: "Do I know **how many times**?" Yes → `for`. "Do I only know **when to stop**?" → `while`. Nested loops are like a clock: the minute hand (inner loop) goes all the way round for each step of the hour hand (outer loop).

COMMON MISTAKE

Indentation decides which loop a line belongs to. In the triangle program, if `print()` were indented under the inner loop, every star would go on its own line.

اردو میں سمجھیں

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ شروع سے آخر تک مکمل چلتا ہے۔ جب دہرانے کی تعداد معلوم ہو تو `for` لوپ، اور جب صرف رکے کی شرط معلوم ہو تو `while` لوپ بہتر ہے۔

EXAM TIP

"Differentiate between for and while loops" (give three points and an example of each) and "How many times will the inner statement run?" (multiply the counts) are typical. For pattern programs, explain what the outer and inner loops each control.

CHECK YOURSELF

1. Predict the output:

```
count = 0
for i in range(3):
    for j in range(2):
        count += 1
    print(count)
```

2. Which loop would you use to keep reading numbers until the user enters 0?

Answers: 1. 6 (3×2). 2. A `while` loop, because the number of inputs is not known in advance.

Next lecture: Practical Activity: Shopping Bill Calculator